Years (Actuals)

- 2013
- 2014
- 2015
- 2016
- 2017
- 2018
- 2019
- 2020
- 2021
- 2022
- 2023
- 2024

Years (Forecasted) + Cases

- 2025
- 2026
- 2027

- Base Case
- Best Case
- Worst Case

Analyst Expectations

International Monetary Fund

FY 2025 - 26	7.3
FY 2026 - 27	6.4

World Bank

FY 2025 - 26	7.4
FY 2026 - 27	6.5

RBI

FY 2025 - 26	7.6
FY 2026 - 27	6.9

Other Rating Agencies

FY 2025 - 26	6.2 - 6.7
FY 2026 - 27	6.6

This analysis has been prepared strictly for educational and informational purposes only. It is based on publicly available data, estimates, and assumptions that may change over time. The content does not constitute financial, investment, legal, or professional advice of any kind. While reasonable care has been taken to ensure accuracy, no guarantee is made regarding completeness or reliability. Any decisions made based on this information are solely at the user's own risk. Readers are strongly encouraged to conduct independent research and consult qualified financial or economic professionals before making any investment or policy-related decisions based on this analysis.

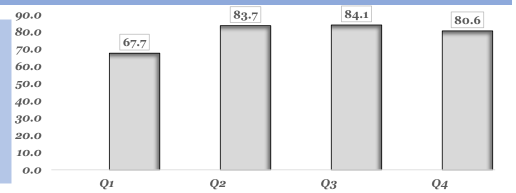
Forecasting of India's GDP

With The Help of Regression Analysis

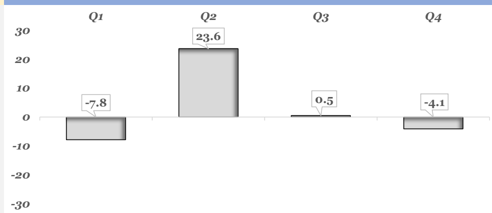
Data Used 2013 - 2025

Actual Annual GDP + Quarterly Chart

316.1



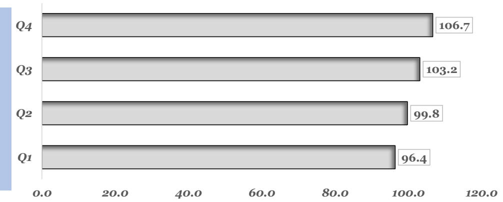
Actual Annual GDP Growth + Quarterly Chart



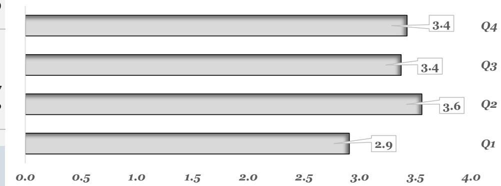
12.2

Forecasted Annual GDP + Quarterly Chart

406.0

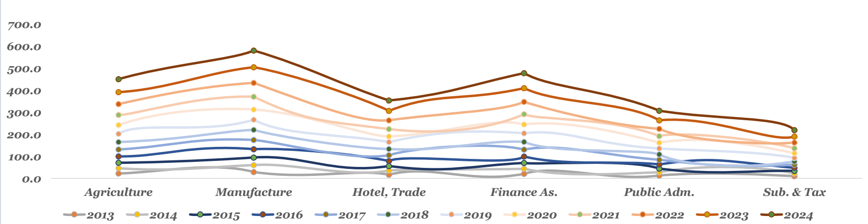


Forecasted Annual GDP Growth + Quarterly Chart



13.3

Production Method - Representation



Sector - Wise GDP

Historical Data

Data Used 2013 - 2025

Sector - Wise GDP (Expenditure Method) + Comparison

Year	PFCE	GFCE	GFCF	CIS	Valuables	Net Exports
2013	64.8	11.6	35.2	1.4	1.6	-3.3503
2014	72.5	13.0	37.5	3.1	2.1	-3.72
2015	81.3	14.4	39.6	2.6	2.0	-3.16
2016	91.3	15.9	43.4	1.4	1.7	-2.72
2017	100.4	18.4	48.2	2.4	2.4	-5.39
2018	112.1	20.5	55.7	3.2	2.3	-7.11
2019	122.5	22.1	57.2	1.4	2.0	-5.18
2020	121.3	23.1	54.3	0.4	2.8	-0.78
2021	143.8	24.7	68.0	2.1	3.9	-6.19
2022	165.3	27.6	84.0	3.0	3.4	-9.61
2023	181.3	31.0	91.7	4.6	4.3	-6.31
2024	203.0	33.0	98.9	5.0	5.0	-7.69

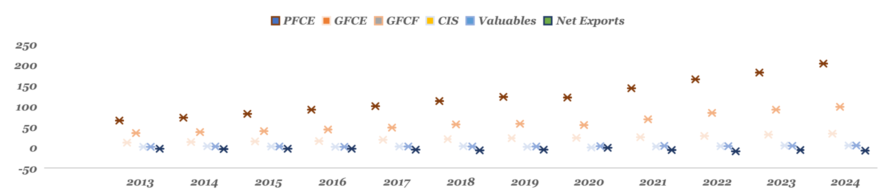
PFCE ; GFCE ; GFCF showing consistent improvement while next exports remain negative all the time

Sector - Wise GDP (Production Method) + Comparison

Year	Agriculture	Manufacture	Hotel, Trade	Finance As.	Public Adm.	Sub. & Tax
2013	22.2	29.0	16.9	20.7	13.0	10.2
2014	24.0	31.4	19.0	23.6	14.9	11.4
2015	25.2	34.7	20.7	26.3	16.6	12.4
2016	28.5	37.7	22.9	29.1	19.0	13.8
2017	31.7	41.9	25.9	31.3	21.4	14.3
2018	34.1	46.1	28.8	35.3	24.2	15.1
2019	37.3	45.9	31.3	38.8	27.1	15.9
2020	40.3	46.6	25.8	40.3	26.3	21.3
2021	45.4	58.1	33.1	46.5	29.7	22.5
2022	49.6	63.2	39.7	56.0	33.6	26.3
2023	54.1	70.9	43.5	62.4	38.4	27.4
2024	59.3	76.0	47.3	68.8	43.5	28.9

Producing sector remains consistent with respect to growth but agriculture sector can be better

Expenditure Method - Representation



India's Net Exports are always negative which might affect or pressurize currency in future

GDP DATA SHEET

Expenditure Method

Particulars (Figures in Lakh Crores)	PFCE	GFCE	GFCF	CIS	Valuables	Exports	Imports	By Bal.	GDP
2013									
Quarter 1	14.2	2.1	7.0	0.4	0.2	6.9	7.3	0.0	23.5
Quarter 2	15.5	2.5	8.1	0.3	0.3	7.1	7.7	1.9	28.1
Quarter 3	19.4	2.9	9.5	0.4	0.6	7.4	8.9	-2.7	28.7
Quarter 4	15.5	4.0	10.5	0.4	0.4	7.1	8.0	-2.1	28.0
2014									
Quarter 1	15.9	2.3	7.5	0.8	0.3	6.9	7.4	-0.2	26.1
Quarter 2	17.4	2.9	8.6	0.6	0.4	7.2	7.8	1.9	31.2
Quarter 3	21.7	3.3	10.1	0.9	0.8	7.4	9.1	-3.5	31.7
Quarter 4	17.4	4.6	11.3	0.8	0.5	7.2	8.1	-2.5	31.1
2015									
Quarter 1	17.9	2.6	7.9	0.7	0.3	6.5	7.0	-0.3	28.6
Quarter 2	19.5	3.2	9.1	0.5	0.4	6.8	7.3	1.8	34.0
Quarter 3	24.4	3.6	10.7	0.8	0.8	7.1	8.5	-4.2	34.7
Quarter 4	19.5	5.0	11.9	0.7	0.5	6.8	7.6	-2.7	34.1
2016									
Quarter 1	20.1	2.9	8.7	0.3	0.3	7.1	7.4	-0.7	31.2
Quarter 2	21.9	3.5	10.0	0.3	0.3	7.4	7.7	3.1	38.7
Quarter 3	27.4	4.0	11.7	0.4	0.7	7.7	9.0	-3.9	38.9
Quarter 4	21.9	5.6	13.0	0.3	0.4	7.4	8.1	-3.5	37.0
2017									
Quarter 1	22.1	3.3	9.6	0.6	0.4	7.7	8.6	-0.2	34.8
Quarter 2	24.1	4.0	11.1	0.5	0.5	8.0	9.0	3.9	43.1
Quarter 3	30.1	4.6	13.0	0.7	1.0	8.4	10.5	-3.9	43.4
Quarter 4	24.1	6.4	14.4	0.6	0.6	8.0	9.4	-3.5	41.4
2018									
Quarter 1	24.7	3.7	11.1	0.8	0.3	9.0	10.3	-0.8	38.5
Quarter 2	26.9	4.5	12.8	0.6	0.5	9.4	10.7	3.6	47.6
Quarter 3	33.6	5.1	15.0	1.0	0.9	9.8	12.5	-5.1	47.8
Quarter 4	26.9	7.2	16.7	0.8	0.6	9.4	11.2	-4.5	45.8
2019									
Quarter 1	26.9	4.0	11.4	0.3	0.3	9.0	9.8	-1.2	41.0
Quarter 2	29.4	4.9	13.2	0.3	0.4	9.4	10.2	3.5	50.7
Quarter 3	36.7	5.5	15.4	0.4	0.8	9.8	12.0	-5.8	50.9
Quarter 4	29.4	7.7	17.2	0.3	0.5	9.4	10.7	-4.8	49.0
2020									
Quarter 1	26.7	4.2	10.9	0.1	0.4	8.9	8.7	-2.8	39.6
Quarter 2	29.1	5.1	12.5	0.1	0.6	9.3	9.1	2.2	49.7
Quarter 3	36.4	5.8	14.6	0.1	1.1	9.6	10.6	-7.3	49.8
Quarter 4	29.1	8.1	16.3	0.1	0.7	9.3	9.5	-6.7	47.3
2021									
Quarter 1	31.6	4.4	13.6	0.5	0.6	12.1	13.0	-1.6	48.3
Quarter 2	34.5	5.4	15.6	0.4	0.8	12.6	13.6	4.3	60.1
Quarter 3	43.1	6.2	18.3	0.6	1.5	13.1	15.9	-6.7	60.4
Quarter 4	34.5	8.7	20.4	0.5	1.0	12.6	14.2	-6.1	57.4
2022									
Quarter 1	36.4	5.0	16.8	0.8	0.5	15.0	16.6	-2.6	55.2
Quarter 2	39.7	6.1	19.3	0.6	0.7	15.6	17.3	3.8	68.4
Quarter 3	49.6	6.9	22.7	0.9	1.4	16.3	20.2	-8.9	68.6
Quarter 4	39.7	9.7	25.2	0.8	0.9	15.6	18.0	-8.2	65.5
2023									
Quarter 1	39.9	5.6	18.3	1.2	0.6	15.5	16.3	-3.0	61.8
Quarter 2	43.5	6.8	21.1	0.9	0.9	16.2	17.0	4.1	76.4
Quarter 3	54.4	7.8	24.7	1.4	1.7	16.8	19.9	-10.1	76.8
Quarter 4	43.5	10.9	27.5	1.2	1.1	16.2	17.7	-9.1	73.4
2024									
Quarter 1	44.7	5.9	19.8	1.2	0.8	16.8	17.9	-3.6	67.7
Quarter 2	48.7	7.3	22.7	1.0	1.0	17.5	18.6	4.1	83.7
Quarter 3	60.9	8.3	26.7	1.5	2.0	18.2	21.7	-11.7	84.1
Quarter 4	48.7	11.6	29.7	1.2	1.3	17.5	19.4	-9.9	80.6
2025									
Quarter 1	48.3	6.3	21.4	1.3	0.9	18.4	19.5	-4.6	72.5
Quarter 2	52.7	7.7	24.6	1.0	1.2	19.2	20.3	3.0	89.1
Quarter 3	65.9	8.8	28.9	1.6	2.3	19.9	23.7	-14.0	89.7

Production Method

Particulars	Agriculture	Manufacture	Hotel, Trading,	Financial Assistance	Public Admin.	GVA	Taxes	Subsidies	GDP
2013									
Quarter 1	4.0	6.7	4.1	5.0	2.3	22.1	1.7	0.3	23.5
Quarter 2	7.1	6.9	4.5	5.2	2.9	26.6	1.9	0.4	28.1
Quarter 3	6.7	8.1	3.7	5.2	3.3	27.0	2.2	0.5	28.7
Quarter 4	4.4	7.2	4.5	5.4	4.6	26.1	2.6	0.8	28.0
2014									
Quarter 1	4.3	7.2	4.6	5.7	2.7	24.5	1.9	0.3	26.1
Quarter 2	7.7	7.5	5.1	5.9	3.3	29.5	2.1	0.4	31.2
Quarter 3	7.2	8.8	4.2	5.9	3.7	29.9	2.4	0.5	31.7
Quarter 4	4.8	7.9	5.1	6.1	5.2	29.1	2.9	0.8	31.1
2015									
Quarter 1	4.5	8.0	5.0	6.3	3.0	26.9	2.0	0.3	28.6
Quarter 2	8.1	8.3	5.5	6.6	3.7	32.1	2.3	0.5	34.0
Quarter 3	7.6	9.7	4.6	6.6	4.2	32.6	2.6	0.6	34.7
Quarter 4	5.0	8.7	5.5	6.8	5.8	31.9	3.1	0.9	34.1
2016									
Quarter 1	5.1	8.7	5.6	7.0	3.4	29.8	2.0	0.6	31.2
Quarter 2	9.1	9.0	6.1	7.3	4.2	35.7	3.6	0.6	38.7
Quarter 3	8.5	10.6	5.1	7.3	4.7	36.2	3.4	0.7	38.9
Quarter 4	5.7	9.4	6.1	7.6	6.6	35.4	2.3	0.6	37.0
2017									
Quarter 1	5.7	9.6	6.3	7.5	3.9	33.0	2.2	0.4	34.8
Quarter 2	10.1	10.1	6.9	7.8	4.7	39.6	4.0	0.5	43.1
Quarter 3	9.5	11.7	5.8	7.8	5.4	40.2	3.7	0.5	43.4
Quarter 4	6.3	10.5	6.9	8.1	7.5	39.3	2.5	0.5	41.4
2018									
Quarter 1	6.1	10.6	7.0	8.5	4.4	36.6	2.4	0.5	38.5
Quarter 2	10.9	11.1	7.7	8.8	5.3	43.8	4.2	0.5	47.6
Quarter 3	10.2	12.9	6.4	8.8	6.1	44.4	4.0	0.6	47.8
Quarter 4	6.8	11.5	7.7	9.2	8.5	43.7	2.6	0.5	45.8
2019									
Quarter 1	6.7	10.6	7.7	9.3	4.9	39.1	2.4	0.5	41.0
Quarter 2	11.9	11.0	8.3	9.7	6.0	46.9	4.3	0.5	50.7
Quarter 3	11.2	12.8	7.0	9.7	6.8	47.5	4.1	0.6	50.9
Quarter 4	7.5	11.5	8.3	10.1	9.5	46.8	2.7	0.6	49.0
2020									
Quarter 1	7.3	10.7	6.3	9.7	4.7	38.7	2.6	1.6	39.6
Quarter 2	12.9	11.2	6.9	10.1	5.8	46.8	4.6	1.7	49.7
Quarter 3	12.1	13.1	5.7	10.1	6.6	47.5	4.3	2.0	49.8
Quarter 4	8.1	11.7	6.9	10.5	9.2	46.3	2.9	1.8	47.3
2021									
Quarter 1	8.2	13.4	8.1	11.2	5.3	46.1	3.2	1.0	48.3
Quarter 2	14.5	13.9	8.8	11.6	6.5	55.4	5.8	1.1	60.1
Quarter 3	13.6	16.3	7.4	11.6	7.4	56.3	5.4	1.2	60.4
Quarter 4	9.1	14.5	8.8	12.1	10.4	54.9	3.6	1.1	57.4
2022									
Quarter 1	8.9	14.5	9.7	13.4	6.0	52.6	3.8	1.2	55.2
Quarter 2	15.9	15.2	10.6	14.0	7.4	63.0	6.7	1.3	68.4
Quarter 3	14.9	17.7	8.8	14.0	8.4	63.8	6.3	1.5	68.6
Quarter 4	9.9	15.8	10.6	14.6	11.8	62.6	4.2	1.3	65.5
2023									
Quarter 1	9.7	16.3	10.6	15.0	6.9	58.6	4.2	0.9	61.8
Quarter 2	17.3	17.0	11.6	15.6	8.4	70.0	7.4	1.0	76.4
Quarter 3	16.2	19.9	9.7	15.6	9.6	71.0	7.0	1.2	76.8
Quarter 4	10.8	17.7	11.6	16.2	13.4	69.8	4.7	1.0	73.4
2024									
Quarter 1	10.7	17.5	11.6	16.5	7.8	64.1	4.5	0.9	67.7
Quarter 2	19.0	18.2	12.6	17.2	9.6	76.6	8.0	0.9	83.7
Quarter 3	17.8	21.3	10.5	17.2	10.9	77.7	7.5	1.1	84.1
Quarter 4	11.9	19.0	12.6	17.9	15.2	76.6	5.0	1.0	80.6
2025									
Quarter 1	10.7	18.8	12.4	18.4	8.9	69.2	4.7	1.4	72.5
Quarter 2	19.0	19.6	13.5	19.1	10.9	82.2	8.4	1.4	89.1
Quarter 3	17.8	22.9	11.3	19.1	12.4	83.5	7.8	1.6	89.7

MACRO DATA SHEET

Actual Value Format | Lag Format

Particulars (Figures in Lakh Crores)	CPI (Inflation)	Repo Rate	Money S.	IIP	Fiscal Deficit	Exports (Net)	Exchange Rate	Crude Oil	CPI (Inflation) Lag	IIP Lag	Repo Lag	Money S. Lag	Exchange R. Lag	Crude Oil Lag
2013														
Quarter 1	10.2	7.5	12.5	1.2	4.8	-0.5	54.5	112.0	--	--	--	--	--	--
Quarter 2	9.7	7.3	11.8	0.8	4.8	-0.5	58.5	103.0	10.2	1.2	7.5	12.5	54.5	112.0
Quarter 3	9.4	7.5	11.2	-0.5	4.8	-1.5	62.5	110.0	9.7	0.8	7.3	11.8	58.5	103.0
Quarter 4	8.3	7.8	10.9	1.0	4.8	-0.8	61.5	109.0	9.4	-0.5	7.5	11.2	62.5	110.0
2014														
Quarter 1	8.2	8.0	10.7	3.5	4.5	-0.6	62.0	108.0	8.3	1.0	7.8	10.9	61.5	109.0
Quarter 2	7.9	8.0	10.5	3.8	4.5	-0.6	60.0	109.0	8.2	3.5	8.0	10.7	62.0	108.0
Quarter 3	6.7	8.0	10.3	2.9	4.5	-1.6	61.5	102.0	7.9	3.8	8.0	10.5	60.0	109.0
Quarter 4	5.6	8.0	10.1	4.2	4.5	-0.9	62.5	76.0	6.7	2.9	8.0	10.3	61.5	102.0
2015														
Quarter 1	5.3	7.8	10.0	3.0	4.1	-0.5	62.5	54.0	5.6	4.2	8.0	10.1	62.5	76.0
Quarter 2	5.8	7.5	9.8	4.1	4.1	-0.5	63.5	62.0	5.3	3.0	7.8	10.0	62.5	54.0
Quarter 3	4.4	7.3	9.6	3.5	4.1	-1.4	65.5	50.0	5.8	4.1	7.5	9.8	63.5	62.0
Quarter 4	5.6	6.8	9.4	2.8	4.1	-0.8	66.5	44.0	4.4	3.5	7.3	9.6	65.5	50.0
2016														
Quarter 1	5.4	6.8	9.2	2.5	3.9	-0.3	67.5	33.0	5.6	2.8	6.8	9.4	66.5	44.0
Quarter 2	5.6	6.5	9.0	6.0	3.9	-0.4	66.5	46.0	5.4	2.5	6.8	9.2	67.5	33.0
Quarter 3	5.0	6.5	8.8	5.2	3.9	-1.4	67.0	46.0	5.6	6.0	6.5	9.0	66.5	46.0
Quarter 4	3.7	6.3	10.5	1.8	3.9	-0.7	68.0	49.0	5.0	5.2	6.5	8.8	67.0	46.0
2017														
Quarter 1	3.7	6.3	9.5	2.7	3.5	-0.9	67.0	53.0	3.7	1.8	6.3	10.5	68.0	49.0
Quarter 2	2.5	6.3	9.2	3.9	3.5	-1.0	64.5	50.0	3.7	2.7	6.3	9.5	67.0	53.0
Quarter 3	3.2	6.0	8.9	4.6	3.5	-2.2	65.0	52.0	2.5	3.9	6.3	9.2	64.5	50.0
Quarter 4	4.5	6.0	8.7	5.2	3.5	-1.3	64.0	60.0	3.2	4.6	6.0	8.9	65.0	52.0
2018														
Quarter 1	4.5	6.0	9.0	5.5	3.5	-1.3	64.5	67.0	4.5	5.2	6.0	8.7	64.0	60.0
Quarter 2	4.9	6.3	9.3	5.3	3.5	-1.3	67.0	74.0	4.5	5.5	6.0	9.0	64.5	67.0
Quarter 3	3.8	6.5	9.6	4.8	3.5	-2.7	71.5	75.0	4.9	5.3	6.3	9.3	67.0	74.0
Quarter 4	2.9	6.5	9.8	3.9	3.5	-1.8	72.0	67.0	3.8	4.8	6.5	9.6	71.5	75.0
2019														
Quarter 1	2.6	6.5	10.0	3.8	3.8	-0.8	71.0	63.0	2.9	3.9	6.5	9.8	72.0	67.0
Quarter 2	3.1	6.3	10.2	2.6	3.8	-0.9	69.5	69.0	2.6	3.8	6.5	10.0	71.0	63.0
Quarter 3	4.0	5.8	10.5	1.5	3.8	-2.2	71.5	62.0	3.1	2.6	6.3	10.2	69.5	69.0
Quarter 4	5.7	5.2	11.0	-0.8	3.8	-1.3	72.5	63.0	4.0	1.5	5.8	10.5	71.5	62.0
2020														
Quarter 1	6.7	5.2	11.5	-3.5	4.6	0.2	72.0	50.0	5.7	-0.8	5.2	11.0	72.5	63.0
Quarter 2	6.8	4.4	12.5	-20.5	6.5	0.2	75.5	29.0	6.7	-3.5	5.2	11.5	72.0	50.0
Quarter 3	6.9	4.0	13.0	-1.2	8.0	-1.0	74.0	43.0	6.8	-20.5	4.4	12.5	75.5	29.0
Quarter 4	6.3	4.0	12.8	1.5	9.2	-0.2	73.5	45.0	6.9	-1.2	4.0	13.0	74.0	43.0
2021														
Quarter 1	5.0	4.0	12.5	10.5	9.2	-0.9	73.0	61.0	6.3	1.5	4.0	12.8	73.5	45.0
Quarter 2	5.6	4.0	11.8	35.0	8.5	-1.0	74.5	69.0	5.0	10.5	4.0	12.5	73.0	61.0
Quarter 3	5.3	4.0	11.2	5.5	7.5	-2.7	74.0	73.0	5.6	35.0	4.0	11.8	74.5	69.0
Quarter 4	5.4	4.0	10.8	2.0	6.7	-1.5	75.5	79.0	5.3	5.5	4.0	11.2	74.0	73.0
2022														
Quarter 1	6.0	4.0	10.5	4.5	6.4	-1.6	74.5	98.0	5.4	2.0	4.0	10.8	75.5	79.0
Quarter 2	7.3	4.4	10.2	12.0	6.4	-1.7	77.5	112.0	6.0	4.5	4.0	10.5	74.5	98.0
Quarter 3	7.0	5.4	9.8	7.0	6.4	-3.9	79.5	101.0	7.3	12.0	4.4	10.2	77.5	112.0
Quarter 4	6.2	6.3	9.5	3.5	6.4	-2.4	82.0	89.0	7.0	7.0	5.4	9.8	79.5	101.0
2023														
Quarter 1	6.4	6.5	9.3	3.5	5.9	-0.8	82.5	82.0	6.2	3.5	6.3	9.5	82.0	89.0
Quarter 2	5.2	6.5	9.0	5.2	5.9	-0.9	82.0	78.0	6.4	3.5	6.5	9.3	82.5	82.0
Quarter 3	6.4	6.5	8.8	6.0	5.9	-3.1	83.0	86.0	5.2	5.2	6.5	9.0	82.0	78.0
Quarter 4	5.5	6.5	8.6	5.0	5.9	-1.6	83.5	83.0	6.4	6.0	6.5	8.8	83.0	86.0
2024														
Quarter 1	5.1	6.5	8.5	5.5	5.8	-1.1	83.0	82.0	5.5	5.0	6.5	8.6	83.5	83.0
Quarter 2	5.0	6.5	8.3	6.0	5.8	-1.1	83.5	84.0	5.1	5.5	6.5	8.5	83.0	82.0
Quarter 3	5.3	6.5	8.2	6.5	5.8	-3.6	83.2	86.0	5.0	6.0	6.5	8.3	83.5	84.0
Quarter 4	5.4	6.5	8.0	5.8	5.8	-1.9	83.5	85.0	5.3	6.5	6.5	8.2	83.2	86.0
2025														
Quarter 1	5.2	6.5	7.9	5.5	5.6	-1.1	83.4	84.0	5.4	5.8	6.5	8.0	83.5	85.0
Quarter 2	5.1	6.5	7.8	5.8	5.6	-1.2	83.6	85.0	5.2	5.5	6.5	7.9	83.4	84.0
Quarter 3	5.0	6.5	7.7	6.0	5.6	-3.8	83.8	86.0	5.1	5.8	6.5	7.8	83.6	85.0

REGRESSION

GDP Growth / GDP Growth Lag (Regression)

Particulars (Figures in Lakh Crores)	GDP Growth	GDP Growth Lag
2013		
Quarter 1	3.4	--
Quarter 2	19.9	3.4
Quarter 3	1.8	19.9
Quarter 4	-2.4	1.8
2014		
Quarter 1	-6.7	-2.4
Quarter 2	19.6	-6.7
Quarter 3	1.8	19.6
Quarter 4	-2.0	1.8
2015		
Quarter 1	-8.3	-2.0
Quarter 2	19.1	-8.3
Quarter 3	1.9	19.1
Quarter 4	-1.6	1.9
2016		
Quarter 1	-8.4	-1.6
Quarter 2	23.9	-8.4
Quarter 3	0.4	23.9
Quarter 4	-4.7	0.4
2017		
Quarter 1	-6.0	-4.7
Quarter 2	23.9	-6.0
Quarter 3	0.5	23.9
Quarter 4	-4.6	0.5
2018		
Quarter 1	-6.8	-4.6
Quarter 2	23.4	-6.8
Quarter 3	0.6	23.4
Quarter 4	-4.2	0.6
2019		
Quarter 1	-10.5	-4.2
Quarter 2	23.7	-10.5
Quarter 3	0.3	23.7
Quarter 4	-3.7	0.3
2020		
Quarter 1	-19.1	-3.7
Quarter 2	25.4	-19.1
Quarter 3	0.3	25.4
Quarter 4	-5.0	0.3
2021		
Quarter 1	2.1	-5.0
Quarter 2	24.4	2.1
Quarter 3	0.5	24.4
Quarter 4	-5.0	0.5
2022		
Quarter 1	-3.8	-5.0
Quarter 2	24.0	-3.8
Quarter 3	0.2	24.0
Quarter 4	-4.5	0.2
2023		
Quarter 1	-5.6	-4.5
Quarter 2	23.7	-5.6
Quarter 3	0.5	23.7
Quarter 4	-4.4	0.5
2024		
Quarter 1	-7.8	-4.4
Quarter 2	23.6	-7.8
Quarter 3	0.5	23.6
Quarter 4	-4.1	0.5
2025		
Quarter 1	-10.1	-4.1
Quarter 2	22.9	-10.1
Quarter 3	0.6	22.9

Regression Statistics

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.285613283
R Square	0.081574948
Adjusted R Square	0.062441092
Standard Error	11.92926825
Observations	50

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	1	606.7110819	606.7110819	4.2633827	0.044367081
Residual	48	6830.757166	142.307441		
Total	49	7437.468248			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
Intercept	4.388421131	1.753560226	2.502577937	0.015787995	0.862651992	7.91419027	0.862651992	7.91419027
X Variable 1	-0.285765113	0.138398712	-2.064796043	0.044367081	-0.564034373	-0.007495852	-0.564034373	-0.007495852

The objective of this study was to examine the relationship between GDP growth and various macroeconomic variables such as CPI (inflation), Index of Industrial Production (IIP), money supply (M3), repo rate, fiscal deficit, exchange rate, crude oil prices, and net exports. The analysis was conducted using regression techniques on quarterly data from 2013 to 2025 to identify statistically significant determinants of economic growth.

The data preparation phase involved converting all variables into a consistent quarterly format. Since some variables were originally available annually, they were distributed across quarters using appropriate assumptions. GDP was calculated using standard approaches and then converted into growth rates to maintain uniformity with other variables such as CPI, IIP, and money supply, which are expressed in percentage terms. This ensured comparability and avoided scale-related distortions in regression analysis.

A key methodological step in this study was the application of **lagged variables**. In macroeconomics, the effect of variables such as inflation, interest rates, or industrial production on GDP is not immediate. Instead, these effects occur with a time lag due to transmission mechanisms such as policy implementation delays, consumption adjustments, and investment decisions. Therefore, each macroeconomic variable was lagged by one period (Lag 1), meaning that the value from the previous quarter was used to explain current GDP growth. This allowed the model to reflect the dynamic nature of economic relationships:

GDP_t = f(X_{t-1})

Initially, multiple regression models were estimated using various combinations of lagged macroeconomic variables. However, the results consistently showed low R^2 values, high p-values, and insignificant F-statistics. This indicated that the selected macroeconomic variables did not have a statistically significant direct impact on GDP growth within the constructed dataset.

There are several reasons for this outcome. First, macroeconomic variables often influence GDP indirectly rather than directly. For example, inflation affects consumption, which in turn affects GDP, while crude oil prices influence production costs and inflation before impacting growth. Second, the impact of these variables may occur over longer time horizons, such as two or three quarters, whereas the model only considered a single-period lag. Third, the dataset included estimated and smoothed values, which can introduce noise and weaken statistical relationships. Additionally, multicollinearity among variables such as CPI and money supply may have reduced their individual explanatory power.

Given the weak performance of macroeconomic variables, the model was refined by introducing an **autoregressive component**, where GDP growth was regressed on its own lagged value:

GDP_t = f(GDP_{t-1})

This approach recognizes that GDP growth exhibits persistence over time, meaning that past growth influences current growth. The regression results for this model showed statistical significance, with a p-value less than 0.05 and a positive adjusted R^2. This confirmed that GDP lag is a meaningful predictor of current GDP growth.

The coefficient of the lagged GDP variable was negative, indicating a **mean-reverting pattern**. This suggests that unusually high growth in one quarter tends to be followed by lower growth in the next, while low growth is often followed by recovery. Such behavior reflects real-world economic cycles, where growth does not remain consistently high or low but fluctuates around a long-term trend.

Although macroeconomic variables such as CPI, IIP, crude oil prices, and exchange rates were found to be statistically insignificant in this model, they are not irrelevant. Their lack of significance may be due to indirect effects, longer lag structures, or limitations in data quality. In more advanced models, such as multi-lag or structural models, these variables may show stronger relationships with GDP.

An important takeaway from this study is the limitation of simple linear regression in capturing the complexity of macroeconomic relationships. The economy is influenced by multiple interconnected factors, and GDP is an aggregate outcome of consumption, investment, government expenditure, and net exports. Therefore, isolating the impact of individual macro variables in a simplified framework can often lead to weak statistical results.

Furthermore, the presence of noise in the data, especially due to estimation and smoothing techniques, reduces the precision of regression coefficients. In real-world applications, economists use more advanced techniques such as Vector Autoregression (VAR), ARIMA models, or structural econometric models to better capture dynamic relationships and multi-period effects.

Another important implication is the role of expectations and behavioral responses in economics. Variables like inflation and interest rates influence decisions of households and firms over time, and their effects may not be immediately observable in GDP figures. This reinforces the importance of lag structures and dynamic modeling in economic analysis.

The study also highlights the importance of model parsimony. Including too many variables can lead to overfitting and multicollinearity, reducing the reliability of results. A simpler model with fewer but statistically significant variables often provides better interpretability and predictive performance.

In conclusion, this analysis demonstrates that GDP growth is better explained by its own past values rather than contemporaneous macroeconomic indicators in a basic regression framework. The use of lagged variables is essential for capturing temporal relationships, and autoregressive models offer a practical and statistically valid approach for short-term economic forecasting.

MODEL CALC. SHEET

Model Calc. With Assumptions

Intercept	**4.388421131**
Coefficient	**-0.285765113**

Particulars (Figures in Cr)	GDP Growth	GDP Lag	Predicted GDP
2013			
Quarter 2	19.9	3.4	3.4
Quarter 3	1.8	19.9	-1.3
Quarter 4	-2.4	1.8	3.9
2014			
Quarter 1	-6.7	-2.4	5.1
Quarter 2	19.6	-6.7	6.3
Quarter 3	1.8	19.6	-1.2
Quarter 4	-2.0	1.8	3.9
2015			
Quarter 1	-8.3	-2.0	4.9
Quarter 2	19.1	-8.3	6.7
Quarter 3	1.9	19.1	-1.1
Quarter 4	-1.6	1.9	3.8
2016			
Quarter 1	-8.4	-1.6	4.8
Quarter 2	23.9	-8.4	6.8
Quarter 3	0.4	23.9	-2.4
Quarter 4	-4.7	0.4	4.3
2017			
Quarter 1	-6.0	-4.7	5.7
Quarter 2	23.9	-6.0	6.1
Quarter 3	0.5	23.9	-2.4
Quarter 4	-4.6	0.5	4.2
2018			
Quarter 1	-6.8	-4.6	5.7
Quarter 2	23.4	-6.8	6.3
Quarter 3	0.6	23.4	-2.3
Quarter 4	-4.2	0.6	4.2
2019			
Quarter 1	-10.5	-4.2	5.6
Quarter 2	23.7	-10.5	7.4
Quarter 3	0.3	23.7	-2.4
Quarter 4	-3.7	0.3	4.3
2020			
Quarter 1	-19.1	-3.7	5.5
Quarter 2	25.4	-19.1	9.9
Quarter 3	0.3	25.4	-2.9
Quarter 4	-5.0	0.3	4.3
2021			
Quarter 1	2.1	-5.0	5.8
Quarter 2	24.4	2.1	3.8
Quarter 3	0.5	24.4	-2.6
Quarter 4	-5.0	0.5	4.2
2022			
Quarter 1	-3.8	-5.0	5.8
Quarter 2	24.0	-3.8	5.5
Quarter 3	0.2	24.0	-2.5
Quarter 4	-4.5	0.2	4.3
2023			
Quarter 1	-5.6	-4.5	5.7
Quarter 2	23.7	-5.6	6.0
Quarter 3	0.5	23.7	-2.4
Quarter 4	-4.4	0.5	4.3
2024			
Quarter 1	-7.8	-4.4	5.6
Quarter 2	23.6	-7.8	6.6
Quarter 3	0.5	23.6	-2.4
Quarter 4	-4.1	0.5	4.3
2025			
Quarter 1	-10.1	-4.1	0.0
Quarter 2	22.9	-10.1	7.3
Quarter 3	0.6	22.9	-2.2

Best Case	Base Case	Worst Case	Intercept	Coefficient
2	0	-2	4.388421131	-0.285765113

Quarter	GDP Growth	Predicted GDP	Actual GDP	Est. GDP
	Lag	*I+(C*GDP Lag)*		*Future GDP Est.*
2025	Best Case			
Quarter 3	0.6	4.2	89.7	--
Quarter 4	4.2	5.2	--	93.4
2026				
Quarter 1	5.2	4.9	--	97.3
Quarter 2	4.9	5.0	--	101.4
Quarter 3	5.0	5.0	--	105.7
Quarter 4	5.0	5.0	--	110.1
2027				
Quarter 1	5.0	5.0	--	114.8
Quarter 2	5.0	5.0	--	119.57
Quarter 3	5.0	5.0	--	124.60
Quarter 4	5.0	5.0	--	129.83
2025	Base Case			
Quarter 3	0.6	4.2	89.7	0.0
Quarter 4	4.2	3.2	--	92.5
2026				
Quarter 1	5.2	2.9	--	96.4
Quarter 2	2.9	3.6	--	99.8
Quarter 3	3.6	3.4	--	103.2
Quarter 4	3.4	3.4	--	106.7
2027				
Quarter 1	3.4	3.4	--	110.3
Quarter 2	3.4	3.4	--	114.1
Quarter 3	3.4	3.4	--	117.9
Quarter 4	3.4	3.4	--	121.9
2025	Worst Case			
Quarter 3	0.6	4.2	89.7	--
Quarter 4	4.2	1.2	--	91.6
2026				
Quarter 1	5.2	0.9	--	95.5
Quarter 2	0.9	2.1	--	98.1
Quarter 3	2.1	1.8	--	100.7
Quarter 4	1.8	1.9	--	103.3
2027				
Quarter 1	1.9	1.9	--	106.0
Quarter 2	1.9	1.9	--	108.8
Quarter 3	1.9	1.9	--	111.6
Quarter 4	1.9	1.9	--	114.5

This model estimates GDP using a linear regression framework based on the relationship between current GDP growth and its lagged values. The model applies an intercept of 4.3884 and a coefficient of -0.28576, indicating a mean-reversion pattern where higher growth in previous periods tends to moderate future growth. Historical quarterly data is used to compute lagged GDP, forming the foundation for predictive calculations.

Future projections are developed under three distinct scenarios—Best Case, Base Case, and Worst Case—each reflecting different economic conditions through adjusted GDP growth assumptions. The Best Case assumes stable and higher growth, the Base Case reflects moderate and realistic trends, while the Worst Case captures potential economic slowdowns. Forecasts are generated iteratively, where each period's output becomes the input for the next, ensuring continuity in projections.

The model maintains simplicity and transparency but has limitations, as it relies on a single explanatory variable and assumes a consistent linear relationship over time. It does not incorporate external macroeconomic factors such as inflation, interest rates, or policy changes. Additionally, the model assumes no structural breaks or unexpected shocks in the economy, which may affect accuracy.

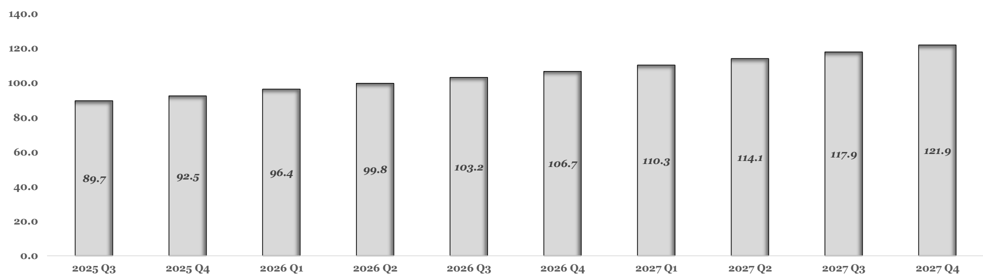
Therefore, it is most effective for scenario analysis, comparative forecasting, and understanding directional economic trends rather than precise long-term predictions.

SCENARIO	Choose Scenario Applicable	2	Best Case	Base Case	Worst Case

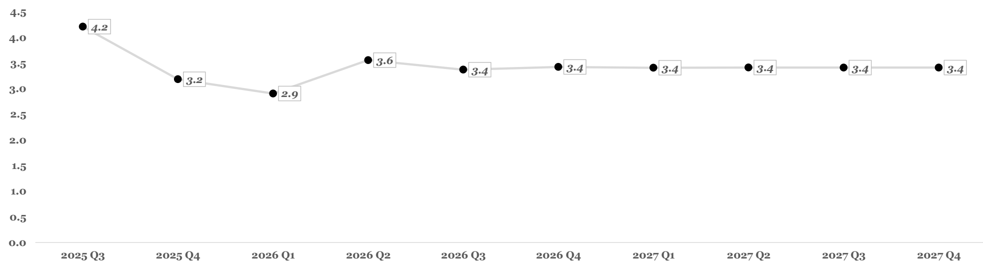
		Best Case	Base Case	Worst Case
Intercept	**4.388421131**	**Best Case**	**Base Case**	**Worst Case**
Coefficient	**-0.285765113**	**2**	**0**	**-2**

Particulars (Figures in Cr)	GDP Growth	Predicted GDP	Actual GDP	Est. GDP
	Lag	*I+(C*GDP Lag)*		*Future GDP Est.*
2025				
Quarter 3	0.6	4.2	89.7	--
Quarter 4	4.2	3.2	--	92.5
2026				
Quarter 1	5.2	2.9	--	96.4
Quarter 2	2.9	3.6	--	99.8
Quarter 3	3.6	3.4	--	103.2
Quarter 4	3.4	3.4	--	106.7
2027				
Quarter 1	3.4	3.4	--	110.3
Quarter 2	3.4	3.4	--	114.1
Quarter 3	3.4	3.4	--	117.9
Quarter 4	3.4	3.4	--	121.9

Base Case Estimated Actual GDP



Base Case Predicted GDP Growth



- Best Case Summary

The Best Case scenario reflects strong and stable economic expansion, with GDP showing a consistent upward trajectory across all forecasted periods. Growth initially adjusts from recent volatility and stabilizes around higher levels (~5%), indicating sustained economic momentum. The projected GDP values increase steadily, supported by strong growth assumptions and positive carry-forward effects. The growth chart highlights minimal fluctuations after stabilization, suggesting a healthy and predictable expansion phase. Overall, this scenario assumes favorable macroeconomic conditions, strong demand, and absence of major disruptions, resulting in optimistic long-term economic performance.

- Base Case Summary

The Base Case scenario represents a moderate and realistic growth path, where GDP increases steadily but at a controlled pace. Growth rates fluctuate in the short term before stabilizing around ~3–3.5%, reflecting balanced economic conditions. The GDP projection chart shows gradual expansion without sharp spikes, indicating sustainable development. The growth trend suggests mild volatility initially, followed by consistency in later periods. This scenario assumes stable macroeconomic factors, no major shocks, and a continuation of current trends. It provides a practical benchmark for expected economic performance under normal conditions.

- Worst Case Summary

The Worst Case scenario captures a slowdown in economic activity, with GDP growth declining initially before recovering slightly and stabilizing at lower levels (~1.9%). The GDP chart shows slower expansion compared to other scenarios, indicating weakened economic momentum. The growth chart highlights early volatility and a dip, followed by limited recovery, suggesting constrained conditions. This scenario assumes adverse macroeconomic factors, reduced demand, or external shocks. Overall, it reflects cautious expectations where growth persists but remains subdued over the forecast period.

The Gross Domestic Product (GDP) data used in this analysis has been collected from reliable and publicly available secondary sources to ensure accuracy and credibility. The primary source of GDP data is the Ministry of Statistics and Programme Implementation (MOSPI), which is responsible for publishing official national income estimates of India. In addition to this, reference has also been made to reports and publications of the Reserve Bank of India (RBI), which provides comprehensive insights into macroeconomic indicators, growth trends, and economic outlook. These sources are widely recognized and ensure that the data used is standardized, consistent, and suitable for academic as well as analytical purposes.

The data collected primarily consists of annual GDP figures at both current and constant prices over a selected time period. These figures were carefully reviewed and organized to maintain uniformity in units, such as ₹ crore or ₹ lakh crore, to avoid inconsistencies during analysis. In certain cases, adjustments and alignment techniques were applied to ensure continuity of data across different years. The use of verified secondary data enhances the reliability of the study and provides a strong foundation for further analysis.

However, the objective of this study is not limited to annual analysis but extends to understanding quarterly economic trends and building a scenario-based forecasting model. Quarterly analysis is crucial as it helps in capturing short-term fluctuations in economic activity, which cannot be observed through annual data alone. Economic performance varies across different quarters due to factors such as seasonal demand, agricultural cycles, festive periods, and variations in government expenditure. Therefore, it becomes necessary to convert annual GDP data into quarterly estimates in order to achieve a more detailed and dynamic analysis.

Since complete and consistent quarterly GDP data may not always be available for all years, a weightage method has been adopted to estimate quarterly values from annual figures. The weightage method is a widely used technique in economic analysis, where the total annual GDP is distributed across four quarters based on assumed proportions that reflect typical economic patterns. The fundamental principle of this method is that the sum of GDP across all four quarters must be equal to the total annual GDP.

In this approach, each quarter is assigned a specific weight representing its share in annual economic activity. These weights are determined based on general economic behavior observed in India. The first quarter, covering April to June, is usually characterized by relatively lower economic activity as it marks the beginning of the financial year. The second quarter, from July to September, shows moderate growth influenced by monsoon-related activities. The third quarter, spanning October to December, typically experiences higher economic activity due to the festive season, which boosts consumption and production. The fourth quarter, from January to March, often records the highest activity as businesses and the government increase spending towards the end of the financial year.

Based on these patterns, appropriate weights are assigned to each quarter, ensuring that their total equals one hundred percent. Once the weights are finalized, the annual GDP is distributed proportionately among the four quarters using a simple calculation method. Each quarter's GDP is obtained by multiplying the annual GDP with the respective weight assigned to that quarter. This process ensures that the total of all quarterly estimates matches the annual GDP, thereby maintaining consistency and accuracy in the model.

The estimated quarterly GDP values derived through this method are then used for further analysis, including the calculation of growth rates, identification of trends, and development of forecasting models. These values play a crucial role in constructing charts, dashboards, and scenario analyses such as best case, base case, and worst case scenarios. The use of quarterly data enhances the depth of analysis and provides a clearer understanding of economic behavior over time.

Despite its usefulness, the weightage method is based on certain assumptions and has some limitations. It assumes that the distribution of economic activity across quarters remains relatively stable over time and does not account for unexpected events such as economic crises, pandemics, or sudden policy changes. As a result, the estimated quarterly figures may not perfectly reflect actual data but serve as a reasonable approximation for analytical purposes.

In conclusion, the methodology adopted in this study involves collecting GDP data from credible sources such as MOSPI and RBI and applying a weightage method to estimate quarterly values from annual data. This approach provides a practical and effective solution for conducting detailed economic analysis in the absence of complete quarterly data. While it involves certain assumptions, it significantly enhances the analytical capability of the model and supports better understanding, visualization, and forecasting of economic trends. Therefore, the use of this method is both appropriate and justified for educational and analytical purposes.